APPENDIX A
Control Equipment Structure

by Sam Herb

Topic Highlights

Definitions
Single-loop Control
Large Systems
Central Computer Control
PLC Control
SCADA Systems
Distributed Control
Hybrid Control
PC Control
Control in the Field

A.1 Introduction

Talking about various controllers is like talking about various vehicles. There is no one best type of transportation; it is all in the application requirements. Sport cars, sedans, and pickup trucks all serve different functions. Tractor trailers offer flexibility, yet they are frequently hauled long distances in railway trains. Yachts and cruise ships have different purposes, as do ocean-going oil tankers, coastal freighters, and ships and barges on large rivers. When it comes to selecting suppliers, it's like automobile buyers—there is a large group who swears by a particular brand, and an equally large group who swears *at* that same brand. Controller and control systems are no different. They all control. There are different applications and different emotions—yes, even for "logical" engineers. For lack of a better way, I will arbitrarily approach this topic from an approximately historical perspective, which may explain the existence and purpose of control equipment.

A.2 Definitions

Before we start, let's define some terms and acronyms often used for controllers to reduce confusion:

SLC – single-loop controller

DAQ – data acquisition

SCADA – supervisory control and data acquisition (pipelines, power distribution, water systems, etc., beyond the plant itself)

DAC – data acquisition and control (some now call this SCADA, but it is within a plant)

DCS – distributed control system

PLC – programmable logic controller

PAC – programmable automation controller (ARC term for the combined functionality of PLCs and PCs; some may call this a version of hybrid controller)

PC Controller – controller uses industrial personal computer (PC), maybe through an SBC

SBC – single board computer (controller); a printed circuit board that contains a complete computer, including processor, memory, I/O, and clock

Hybrid controller – architectures that blend benefits of both PLCs and DCSs; generally "small-scale DCSs"; the term is defined differently by different people.

CIF – control in the field; the control loop resides across a fieldbus only, not within a controller

A.3 Single-loop Control

More than your home furnace control, which is relay on-off control, these modulating controllers were initially for operating valves. They were originally pneumatic/mechanical, then electromechanical, then electric, then electronic (vacuum tube, then transistor), and are now microprocessor-based. This later version makes some networking possible, as well as some more sophisticated control, approaching what is sometimes called "hybrid control."

Because the mechanism of the early controllers provided the motion of the pointer and pen of the indicator and recorder, the "faceplate" was integral to the control, and the many instrument companies called the entire assembly "the controller" (Figure A-1). This later influenced the instrument companies' approach to their respective "distributed control systems."

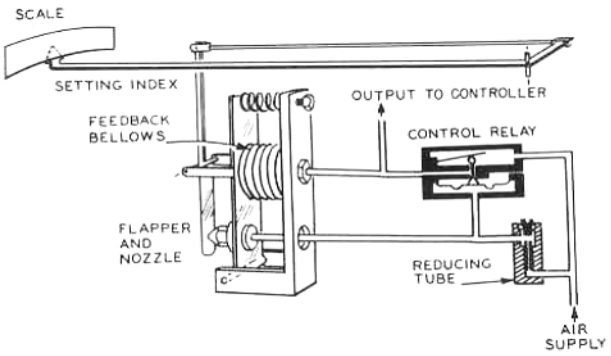


Figure A-1: Measurement and Control Mechanism was One with the Indicator and Recorder

Another design circumstance was the three modes of control as covered in the chapter entitled "Continuous Control," were determined by the interaction of the mechanism (Figure A-2). It did not matter if you had a restrictor, a capacity, or a needle valve, or if you later had a resistor, a capacitor, and a vacuum tube (voltage valve)—or transistor (current valve)—the technology was essentially the same.

To change from a proportional (P) controller, to a proportional-integral (PI) or a proportional-integral-derivative (PID) type controller, you had to change controllers, or at least change the chassis. If you wanted to change strategies, or even tune differently, you had to go out and buy more stuff. Later electronic versions still required different circuit boards. There was no way to merely turn off one of the modes. This is one of the reasons why so many automatic controllers were run in manual mode (some say 85–90%).

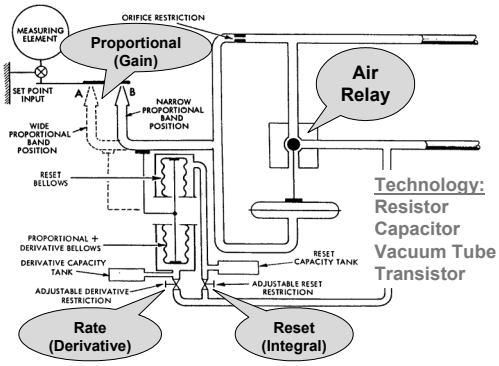


Figure A-2: Interactive Mechanism, Pneumatic or Its Electronic Counterpart, Restricted Control Action Flexibility

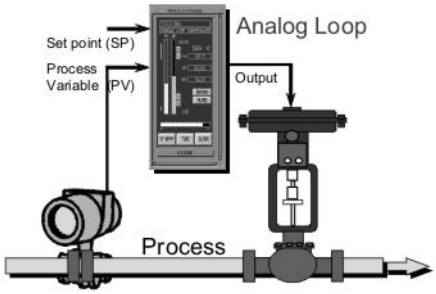


Figure A-3: Traditional Single-loop Control

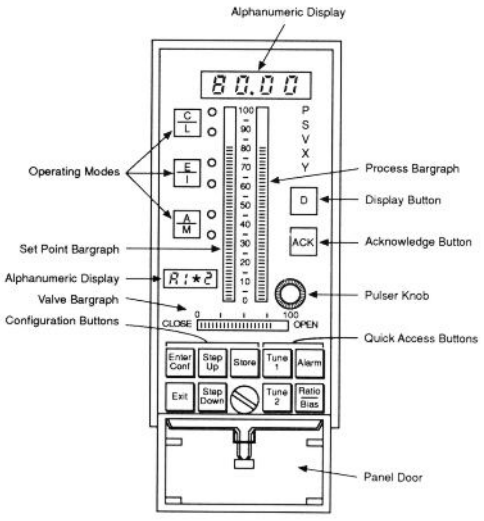


Figure A-4: Microprocessors Made Powerful SLC Possible

Until the microprocessor! Unlike the vacuum tube or the transistor, the microprocessor was not a "valve" but a "counter." This allowed a change in how the control algorithm worked, and permitted the so-called parallel algorithm shown in "Continuous Control." It also gave more choices in tuning

loops. In addition, the microprocessor made more involved control strategies practical. It allowed plants to operate closer to their potential. Microprocessors led to the flexibilities of the distributed control systems and the hybrid controllers.

Single-loop controllers are a very economical way to control simple stand-alone unit processes such as dryers, small furnaces, and small boilers. Because of advances in microprocessors, they can readily be used in small networks as well as team with PLCs, where more sophisticated continuous control is needed along with larger amounts of discrete actions.

A.4 Large Systems

When plants became very large and more complex, the challenge was for the operator to keep track of all the operations and coordinate the many individual unit processes. This meant the many individual single control loops required someone, or several people, to "tour the plant" and record all the data on a clipboard to make a decision (Figure A-5). Then another tour was needed to adjust all the valves and drives. This took time and limited effective plant production.

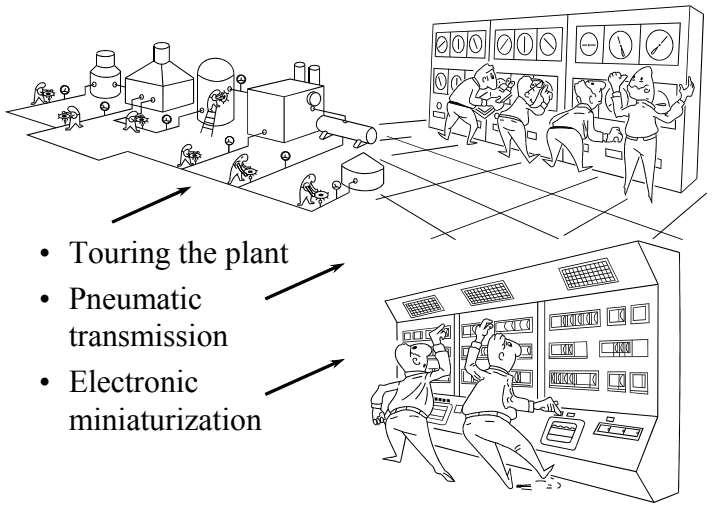


Figure A-5: Limitations of a Single-Loop Control in Large Plants

Following World War II, electronic controls became more rugged and practical for industrial environments. More measurements were becoming possible because the cost of sensors was coming down. This also allowed the industry to develop newer types of sensors for measuring parameters not previously measurable. Further, it was becoming possible to continuously measure more parameters online, rather than tediously take laboratory samples and wait for results.

The size of these new electronic controllers was smaller, so more of them could fit on a panel, in a smaller area. All of this led to a more complex control room, as well as the need to bring more wires to that location. This presented information management problems to the operators and the logistics challenge of signal management to the instrument engineer.

As changes in technology decreased the price of computers, their use became more common in large and complex facilities. This also allowed the further development of the single, centralized control room.

A.5 Central Computer Control

Large central computers were important for large plants and power utilities. While these computers were now able to cope with the new data, most computers were designed for transactional businesses rather than navigational process control. During the 1960s and 1970s, two types of computers emerged for process control:

- Direct digital control
- Digitally directed analog control, more popularly called Supervisory Control

The central control room concept gave a much better picture of the overall plant operation, but when all the distant portions of the plant were connected to this one room, the cost was high because of the following factors:

- Many control cable runs, wire trays, and handling devices
- More complex engineering design
- Craft labor of installing lines and making terminations
- Problems making changes in the control strategy

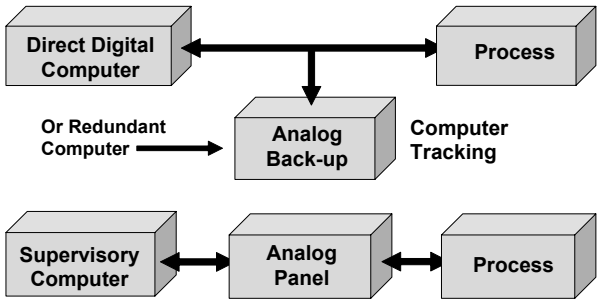


Figure A-6: Central Computer with Analog Backup

A serious problem also presented itself—a failure in the computer could shut down the entire plant! To overcome this limitation, backup controllers were often used with the computer system. As a result, companies frequently had to duplicate control systems (which meant buying two sets of controls for everything) to make systems reliable. This redundancy often required analog instruments to keep the plant running and made changes to control strategy more difficult.

Centralized computer control and monitoring had several advantages and disadvantages:

- On the plus side: there was a more organized central view of the operations; control strategies became more flexible; alarms became much more flexible and effective; and there was increased ability to have meaningful history of events.

- On the negative side: there was a large amount of wiring; there was considerable risk to the plant; and it was not very "scalable," meaning you could not scale up a little bit more without reprogramming the whole computer. All of these negatives cost money.

Computer-like control has become more necessary, however, because as each industry matures, it needs to optimize its processing methods. Cost of raw materials, cost of waste, cost of pollution, cost of compliance to government regulations—all are growing factors in the efficiencies of operations.

The technology of the microprocessor arrived and affected the controls industry for both process control with the DCS and factory automation with the PLC.

A.6 PLC Control

As the acceptance of transistors increased, the "box of relays" grew to be programmable logic controllers. They were primarily used for discrete control in factory automation, and at one time 70% of all PLCs were used in automobile manufacturing. PLCs were designed to be stand alone, and essentially had start-and-stop operator interfaces and little need for communication networks. Each box was configured independently. They provided very fast scanning circuitry for robotic control and numeric control machines for manufacturing parts. Designed to perform in a work cell (workstation for one manufacturing employee), common practice was to have one PLC per function. Work cells in manufacturing are comparable to a unit process in a continuous control plant for such functions as burner management and reactor control.

The chapters "Discrete Input and Output Devices and General Manufacturing Measurements" and "Discrete and Sequence Control" provide more information about PLCs. A large advantage to PLCs has been their low unit price, because they have traditionally been sold as commodities. PLCs are usually installed by local third-party system integrators, and they have usually been configured in Ladder Logic (North America) or Boolean (Europe). Today most have migrated to the IEC 61131-3 "standard," using the five languages that make it easier to configure more sophisticated functions. There is more information on this topic in the chapter.

Another advantage of PLCs, at least in the factory-automation world, is the user's ability to choose an operator station (human-machine interface [HMI]) from a wide selection of different suppliers. For process control, the PLC supplier is likely to provide an HMI of its own design for better compatibility to the greater sophistication needed for alarm management or batch processing.

PLCs became very versatile for many applications such as packaging lines, which led them into food and pharmaceutical applications. This in turn led to the need for more sophisticated control and some single-loop functions. Companies demanded more sophisticated continuous control from PLC suppliers who have headed into the batch markets. The legacy of configuring each controller independently persists (Figure A-7). Most factory-automation applications did not need redundancy, so this has not been a strength of PLCs of the past. Sophisticated redundancy is often a requirement in most process control applications, and those PLC suppliers moving towards those market spaces have begun meeting the challenge.

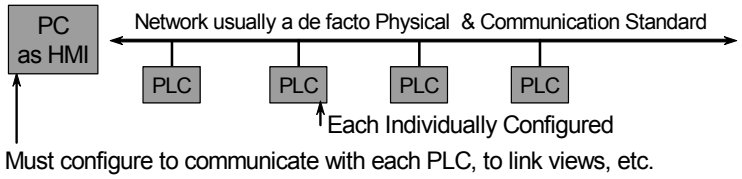


Figure A-7: Typical PLC System Configuration

As PLC applications became more complex, their suppliers have been developing appropriate networks and graphic operator interfaces.

A.7 SCADA Systems

Unfortunately, when suppliers of HMIs began to refer their lash-up of PCs with PLCs as "SCADA" systems, confusion arose in the industry. This is in part because of the simple definition of this acronym for supervisory control and data acquisition. The original reference for systems of PCs with PLCs within a plant was data acquisition and control (DAC).

Traditional SCADA, which has been in existence for more than a half century, has always been used to transmit data over long distances. Common uses include power transmission systems (transmission and distribution), oil and gas pipelines, and water distribution systems. SCADA can be used in any situation where monitoring and control is done over large distances.

A system is made up of a master termination unit (MTU) and one or more remote termination units. The MTU will likely be in the control room, connected to the many functions needed there (Figure A-8). Beyond large control systems, there can be not only operators, maintenance folks, engineers and managers, but also accountants, government regulatory record keeping, and the many functions of custody transfer, depending upon the application.

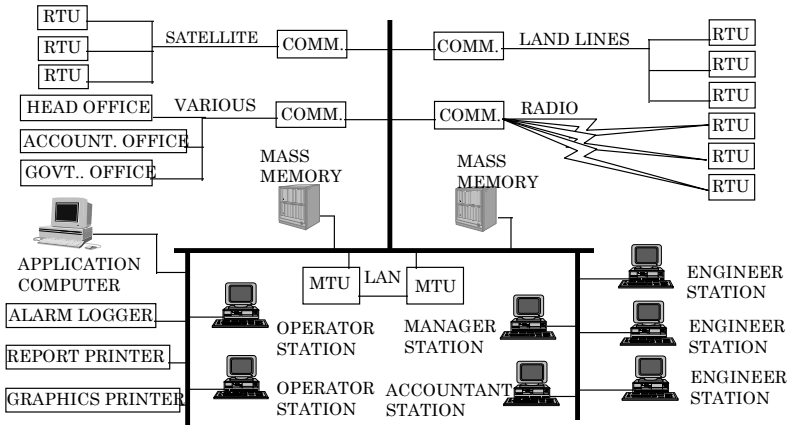


Figure A-8: SCADA is for Applications Over Long Distances

Systems transmit over many kinds of media such as landlines, phone systems (wire or wireless), microwave, radio, and satellite. Because of the distances, any process control done remotely must carefully be stand-alone control because of the time lag from distances and sharing of communications (scanning many RTUs). This is an essential feature, requiring specific communication techniques to ensure the integrity of information and direction given for any control action.

Also because of developments in microprocessor technology, RTUs today can come with many more functions than merely remotely reporting values and turning switches on and off as their earlier cousins did. There are different levels of "smart" RTUs, just as there are different and expanding capabilities with "smart" transmitters and end elements. Some very sophisticated remote stand-alone controllers can be added to the RTU functions.

A.8 Distributed Control

The advent of distributed control came with the useful capabilities of emerging video technologies to display data and to allow an operator to initiate control actions "through the video" as well. The advantage of the central control room was it provided centralized *information* without having all the processing in one vulnerable location, thereby distributing the risk.

Distributed control also saved the cost and complexity of wiring by sending a digital signal through a single cable, which was used as a communication network (data highway) connecting the diverse portions of the plant. The magic of sending many signals through a single wire is an old technology, called the telegraph. The use of Morse code was really digitally communicating analog values (in contrast to analog voice-based information, like radio). There is more on this topic in the chapter "Digital Communications."

This new architecture permitted a *functional* distribution of the tasks among many processors, reducing the risk of everything failing at once. As capabilities for reducing ground loops emerged, it was also possible to allow *physical* distribution as well. These critical features began to open up many new possibilities for tying central information to local control in those plants where this is important.

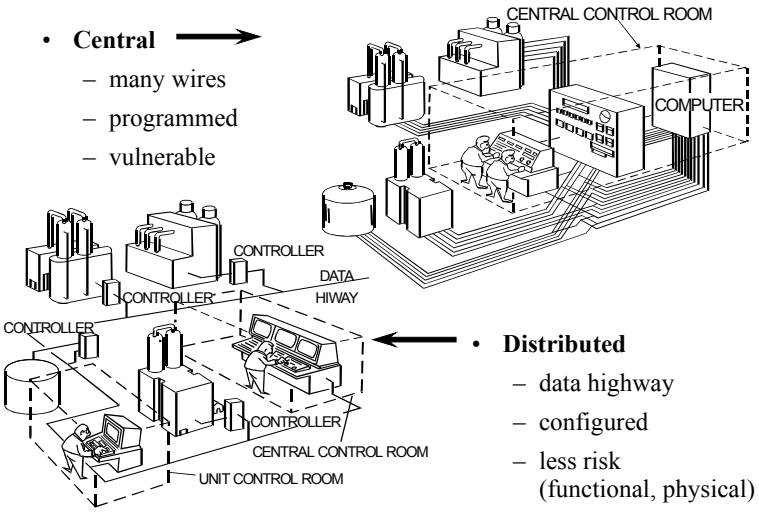


Figure A-9: DCS Saved More Than Wires

The central control room view of plant operation gives operators a single window to the process. Now operators no longer have to tour the plant. They can literally "let their fingers do the walking" as they call up each controller or group of controllers on a screen to check the progress of the process. If needed, they can easily make set point and output changes from the keyboard, as well as respond to any alarms if a process is "off normal."

Furthermore, if a plant process requires it, there also can be several operator stations along this network. A local operator station can be positioned in a specific portion of the plant, either working from the same data highway, or wired directly to a cluster of some loops of control.

On the plus side, distributed control and monitoring meant shorter wiring runs, no wires between controllers and control room, less risk of failure, and a more scalable system, if you wished to gradually grow the system without much cost of replacement.

On the negative side, these distributed control systems still must wire sensors and final elements to control cabinets, and interconnection between different vendors' components presented no small difficulty. This is the realm being addressed by digital I/O—called fieldbus, which is covered in "Digital Communications."

Remember the comment about the instrument companies looking at controllers as including the "faceplate"? Unlike PLC suppliers who needed only a "box" to do a specific function, the DCS suppliers were all instrument companies with the philosophy that the operator interface was integral to the

function of process control. As a result, the control strategies are configured from the workstation, which was designed integral to the controllers (Figure A-10). This enables a single database and a more integrated operator interface for more sophisticated alarm management and other operator functions. Simplistically, operator actions in factory automation were traditionally "on-off," whereas in process control, the operator turned a valve while watching a gauge—the action was more "adjust until."

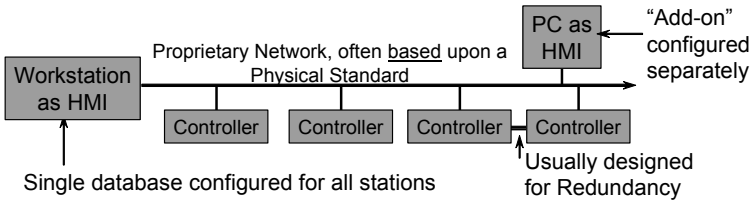


Figure A-10: Typical DCS Configuration

Another distinction was that PLCs had traditionally not needed redundancy or sophisticated communication. DCSs, however, usually did need sophisticated redundancy, and so-called "real-time" communication. No computer can do "real-time," but at best has to act in "real-enough-time," because the process is actively "flowing." To achieve this, the instrument suppliers had to design controller, workstations, and communication networks as a single package. No standard communication method was fast enough, so they each modified the standards to achieve the "real-enough-time" networks to match their individual controller and workstation designs.

DCSs originally used UNIX-based workstations, often very proprietary, to allow for the "real-enough-time" seriously needed, as well as for the unique navigational process functions not accounted for in discrete-based transactional "business machines." As companies began using Microsoft throughout their plants, they began demanding PC-based workstations. This presented several significant challenges to the DCS suppliers, which they began to overcome because PCs were also transaction based.

A.9 Hybrid Control

Wouldn't it be nice to have the cost advantage and simplicity of the PLC combined with the sophistication and scalability of the DCS? Wouldn't it be nice to have the power and sophistication of DCS control on a small scale? Thanks to the microprocessor, these capabilities are emerging. What we haven't mentioned yet was the emerging capabilities of combining a few process control loops and some discrete interlocking that would be helpful for those small-to-midsized industries that have stand-alone unit processes, such as heat treating plants and some food industry functions.

A new market space opened to provide improved production capabilities for these smaller plants, which traditionally had to "kludge together" some sort of control system with a very limited budget. Having a decent answer for these small plants is so new that a perfect definition is wanting. Now for the big question: What is a hybrid system? It depends upon whom you ask. If you look at the many articles in the trade journals and the marketing words by the various vendors, you will begin to realize there are many different understandings of the phrase.

That is why there is some confusion—even to the point where one control magazine suggested the whole idea is really a hoax by the instrument companies to sell more control systems! You must realize that in the early DCS days of the 1970s and 1980s, there were still DCSs that could not do discrete actions, or at least not very well. PLCs at that time were notable for their poor implementation of analog control, especially multiloop control of even low sophistication. Clearly *both* worlds have been

moving towards each other, but both typically bring the baggage of their legacies even in their newer designs. That is why there are still several definitions, which *almost* sound alike, but are somewhat different.

Definitions include:

- Industry (ARC Research, Figure A-11)
- Input and output capability
  - Analog + Discrete I/O
- Function (batch capability)
- Architecture
  - Advantages of both PLCs and DCSs
  - Few disadvantages of either (hopefully)

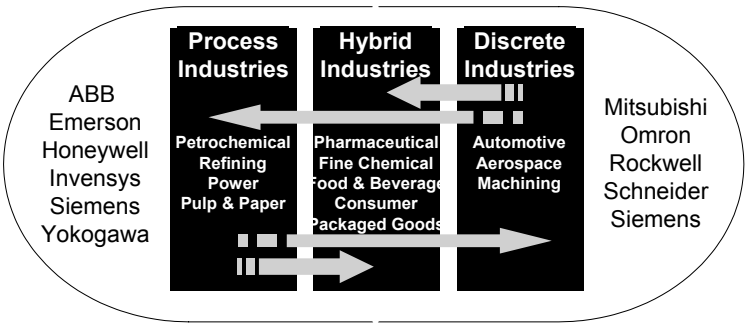


Figure A-11: Hybrid Defined by Industries (From ARC Advisory Group Strategies, June 2001)

Hybrid control first appeared (by name) in the early 1990s with a DCS that allowed users to add any supplier's UNIX- or PC-based workstation to the control system. It uniquely allowed the controllers to be configured as a system, and diagnostics were not limited to each of the "boxes," but reported on the system as a whole, like a DCS. By now "out-of-the-box" workstations could be used, especially if a plant had standardized with a particular workstation supplier. Nevertheless, there was still the issue of needing navigational-type process actions, especially for alarm handling requirements and process batch-like functions.

By the turn of the millennium most suppliers offered PC-based workstations, so this has become the standard. It added to the commercial-off-the-shelf (COTS) needs of the industry. Nevertheless, to achieve the needed process functions, most suppliers also offer their own modified software for these PCs and recommend industrial-grade hardware to survive the harsh conditions of the industries served. This is true of the instrument suppliers and the PLC suppliers who have entered the so-called "hybrid industries" markets.

ARC, which defined "hybrids" as an industry category (Figure A-11), began using the term process automation controller (PAC) to describe the combined functionality of PLCs and PCs. A PAC could include equipment from PLC suppliers, but might be stretched to include equipment from suppliers of "small DCSs . . . and perhaps could also include those who didn't exactly fit either supplier category." Also notice that Figures A-7, A-10, and A-12 look nearly alike, except for the words used. The distinctions are becoming much more subtle. Like most things in this business, begin with the application

and functions you need rather than the labels when buying a control system. My recommendation? Start with the simplest answer for your needs and build upwards as you discover new requirements.

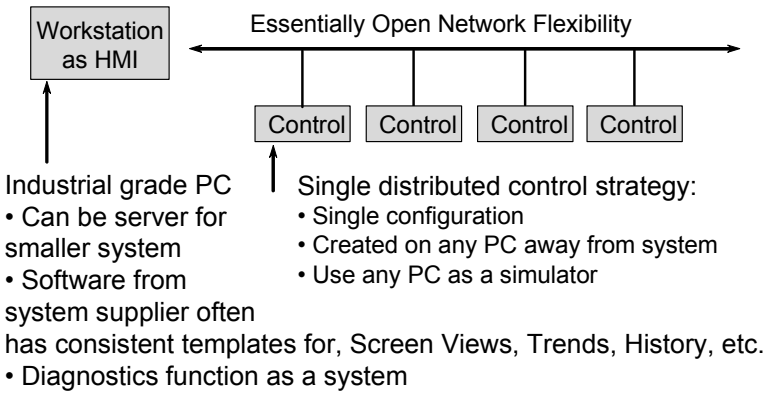


Figure A-12: Hybrid System Architecture Blends PLC Simplicity with DCS Sophistication

A.10 PC Control

Not to be overlooked is the realm of PC controllers. Microsoft technology—and the ability to take advantage of the inexpensive technology—has been hovering over the process control world ever since PCs became so popular. Until recently PC controllers have been limited by the vulnerability of delicate moving components, not practical in many of the harsh environments of the process industries. Nevertheless, this has become a Microsoft world.

The next generation of controllers will be influenced significantly by Microsoft. The trend will be towards off-the-shelf hardware, and away from proprietary systems. Value added by controls vendors will be to the software, which will not only provide control capability, but will still have to overcome the obstacles to "real" real-time. There will also be a significant need for standards in many places for this to be fully realized. This is happening more with PLCs giving way to "soft logic" in factory-automation applications, but some process applications are also suitable. ARC Advisory Group refers to this as Open Control Software (OCS).

Although PC control has the potential to "liberate" the user from being held hostage by a single supplier, there are some remaining considerations about robustness. "Industrial strength" for processes is far more severe than the factory floor where some PC controllers have replaced PLCs. Unlike factory automation, which usually shares the environment with people, process control often occurs in corrosive atmospheres dangerous to people, where vibration is frequently significant. Often, the PC's general-purpose operating system is not stable enough for control. PC-controlled installations are forced to handle system crashes and unplanned rebooting.

As the control and automation market continues to evolve, technologies developed and honed by PC-based control suppliers are finding their way into a diverse range of platforms such as Compact PCI, VME, PC/104, and custom single-board computer (SBC).

A.11 Control in the Field

We complete the circle, which is made possible through the development of digital fieldbus communications, particularly Foundation Fieldbus. Foundation Fieldbus (see "Industrial Networks") provides control function blocks, allowing a complete control loop to exist among the transmitters and end ele-

ments without a specific controller "box" (Figure A-13). With this architecture, a PC can be placed on the fieldbus itself and used as a workstation. The neat part is the field devices can come from different suppliers, because no matter the supplier, the function blocks are written to the same standard so they readily communicate with each other. Nevertheless, the actual content of the function blocks can have unique characteristics, allowing suppliers to differentiate their products.

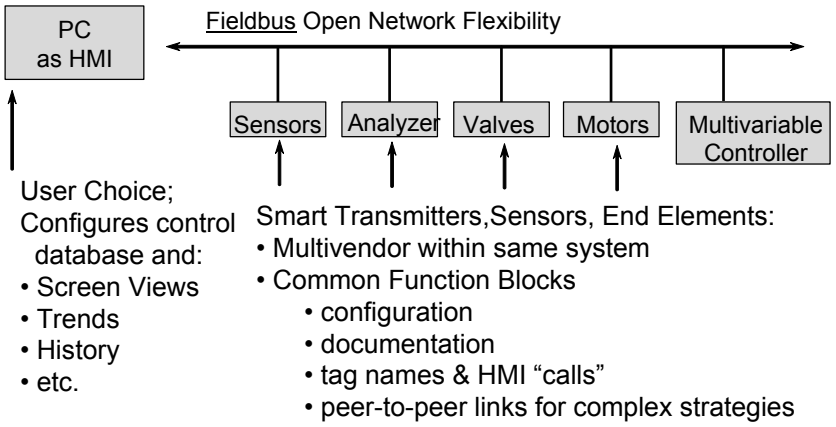


Figure A-13: Foundation Fieldbus Itself can be the Controller

This capability can be very useful in small stand-alone applications. It also allows users to incorporate many separate stand-alone systems into a larger system, with access from a central control room. Fragmenting the control hardware may be counterproductive for more coordinated or multivariable control schemes. The arrangement, however, does not preclude the use of multivariable controllers for more complex control strategies. At present, the limitation is that only a few devices (about six) can exist on the same segment of the bus, with no peer-to-peer communication to other segments. The arrangement may not be practical for more intricate batch applications.

Control strategies can be developed among several transmitters and end elements as shown in Figure A-14. Although some have concerns that the vulnerability of "inner signals" of controllers are at the mercy of the field network, the robustness of installed applications seems to be placing those fears to rest.

Some valve companies have released designs that have placed flow measurement across the valve along with temperature measurement, feeding controller chips and providing a signal directly to the valve actuator—all inside the bonnet of the valve.

A.12 What Next?

Of course new technologies will continually appear to overcome limitations, but I doubt any one of the types covered will fully disappear. Each has some very strong advantages depending upon the application. Remember, it is always the application, and the functions you need, *not* the labels. The good news is that today the *choices* are growing, and generally the prices are coming down, especially when you look at total cost of ownership.

A.13 References

Herb, Samuel M. *Understanding Distributed Processor Systems for Control.* ISA, 1999.

Boyer, Stuart A. *SCADA: Supervisory Control and Data Acquisition, 3rd edition*. ISA, 2004.

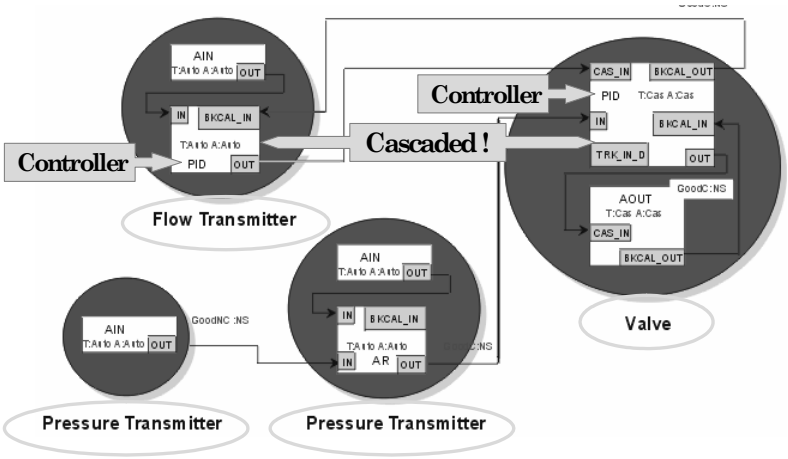


Figure A-14: Control Strategy Fully "in the field"

For further information on the evolution of control see:

Feeley, J., et al. "100 Years of Process Automation." *Control Magazine*. (Vol. XII, No. 12), December 1999 (Special Issue).

Standards

ANSI/ISA-50 Series, Parts 2-6 - *Fieldbus Standard for Use in Industrial Control Systems*.

IEC 61158 Series, Parts 1-6 - *Digital Data Communications for Measurement and Control – Fieldbus for Use in Industrial Control Systems*.

About the Author

Samuel M. "Sam" Herb, PE, is currently with Invensys Process Systems. Herb has worked with Honeywell, Leeds & Northrup, Moore Process Automation, and Siemens during his career. He has worked in the utility and process controls industries for nearly five decades, with various roles in marketing, business development, marketing communications, control applications, product management, systems engineering, project management, product evaluation, technical publications, and education.

Herb holds a BS/EE from Drexel University, is a senior life member of ISA, and is the author of dozens of journal articles for a variety of technical publications, seminars, and courses in instrumentation. In addition, he consulted on several ISA videotape series, authored chapters in the Practical Guide Series on Continuous Control, authored the textbook *Understanding Distributed Processor Systems for Control,* and developed five online courses and CDs on control systems. Herb received the ISA Eagle and Distinguished Society service awards and the Donald P. Eckman award for education.